

Solana On/Off Ramp Audit Summary

Prepared by Cyfrin

Version 2.0

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Securitize on Solana On/Off Ramp. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated February 19, 2026, was delivered to Securitize, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

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5 Audit Overview

Over the course of 13 days, the Cyfrin team conducted an audit on the Solana On/Off Ramp code provided by Securitize. The findings consist of 1 Critical, 1 Medium & 9 Low severity issues with the remainder being informational.

Summary

Project Name	Solana On/Off Ramp
Repository	bc-solana-on-off-ramp-sc
Commit	c9f35ec13465...
Audit Timeline	Feb 3rd - Feb 19th, 2026
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	1	1	0
High Risk	0	0	0
Medium Risk	1	1	0
Low Risk	9	6	3
Informational	6	4	2
Gas Optimizations	0	0	0
Total	17	12	5

6 Audit Scope

The Solana Off-Ramp component was previously audited by Cyfrin; the initial focus of this audit was the Solana On-Ramp.

During the audit the Off-Ramp component received some additional changes, hence the audit was extended to also audit the new changes to the Off-Ramp component. The final scope was limited to:

New on-ramp component:

```
programs/securitize-on-ramp/src/instructions/admin/add_operator.rs
programs/securitize-on-ramp/src/instructions/admin/change_admin.rs
programs/securitize-on-ramp/src/instructions/admin/initialize.rs
programs/securitize-on-ramp/src/instructions/admin/pause.rs
programs/securitize-on-ramp/src/instructions/admin/revoke_operator.rs
programs/securitize-on-ramp/src/instructions/admin/toggle_investor_subscription.rs
programs/securitize-on-ramp/src/instructions/admin/toggle_two_step_transfer.rs
programs/securitize-on-ramp/src/instructions/admin/unpause.rs
programs/securitize-on-ramp/src/instructions/admin/update_asset_provider.rs
programs/securitize-on-ramp/src/instructions/admin/update_fee_manager.rs
programs/securitize-on-ramp/src/instructions/admin/update_min_subscription_amount.rs
programs/securitize-on-ramp/src/instructions/admin/update_nav_provider.rs
programs/securitize-on-ramp/src/instructions/investor/subscribe_ds_token.rs
programs/securitize-on-ramp/src/instructions/investor/swap_ds_token.rs
programs/securitize-on-ramp/src/instructions/operator/swap_spl_token.rs
programs/securitize-on-ramp/src/states/asset_provider/allowance_asset_provider.rs
programs/securitize-on-ramp/src/states/asset_provider/asset_provider_core.rs
programs/securitize-on-ramp/src/states/asset_provider/minting_asset_provider.rs
programs/securitize-on-ramp/src/states/fee_manager/fee_manager_core.rs
programs/securitize-on-ramp/src/states/fee_manager/mpbs_fee_manager.rs
programs/securitize-on-ramp/src/states/nav_provider/amm_nav_provider.rs
programs/securitize-on-ramp/src/states/nav_provider/mod.rs
programs/securitize-on-ramp/src/states/nav_provider/nav_provider_core.rs
programs/securitize-on-ramp/src/states/nav_provider/standard_nav_provider.rs
programs/securitize-on-ramp/src/states/on_ramp_counter.rs
programs/securitize-on-ramp/src/states/on_ramp_state.rs
programs/securitize-on-ramp/src/states/token_type.rs
programs/securitize-on-ramp/src/utils/build_issue_tokens_cpi_data.rs
programs/securitize-on-ramp/src/utils/get_authority_signer.rs
programs/securitize-on-ramp/src/utils/get_transfer_hook_program_id.rs
programs/securitize-on-ramp/src/utils/swap_logic.rs
programs/securitize-on-ramp/src/utils/token_calculator.rs
programs/securitize-on-ramp/src/utils/transfer_maybe_hook.rs
programs/securitize-on-ramp/src/constants.rs
programs/securitize-on-ramp/src/lib.rs
```

Off-ramp changes from the last audit commit 30362cf3d6b349cad72134f843808464d7477502 to the latest commit 401146aed55b708d27982e70f4db1b2866ed37ef (compare); smart contracts only, not including `off-ramp-cli/sdk` etc.

Modified files (excluding events, errors and comment-only changes):

```
programs/securitize-off-ramp/src/constants.rs
programs/securitize-off-ramp/src/instructions/admin/initialize.rs
programs/securitize-off-ramp/src/lib.rs
programs/securitize-off-ramp/src/states/off_ramp_state.rs
programs/securitize-off-ramp/src/utils/get_transfer_hook_program_id.rs
programs/securitize-off-ramp/src/utils/transfer_maybe_hook.rs
```

New files:

```
programs/securitize-off-ramp/src/instructions/admin/add_operator.rs
programs/securitize-off-ramp/src/instructions/admin/revoke_operator.rs
programs/securitize-off-ramp/src/instructions/admin/toggle_two_step_transfer.rs
programs/securitize-off-ramp/src/instructions/admin/update_asset_policy.rs
programs/securitize-off-ramp/src/instructions/admin/update_fee_manager.rs
programs/securitize-off-ramp/src/instructions/admin/update_liquidity_provider.rs
programs/securitize-off-ramp/src/instructions/admin/update_nav_provider.rs
programs/securitize-off-ramp/src/instructions/common/available_liquidity.rs
programs/securitize-off-ramp/src/instructions/common/calculate_liquidity_token_amount.rs
programs/securitize-off-ramp/src/instructions/operator/redeem_spl_token.rs
programs/securitize-off-ramp/src/instructions/user/redeem_ds_token.rs
programs/securitize-off-ramp/src/states/asset_policy.rs
programs/securitize-off-ramp/src/states/countries_restriction.rs
programs/securitize-off-ramp/src/states/fee_manager/fee_manager_core.rs
programs/securitize-off-ramp/src/states/fee_manager/mpbs_fee_manager.rs
programs/securitize-off-ramp/src/states/genesis_hash.rs
programs/securitize-off-ramp/src/states/liquidity_provider/allowance_liquidity_provider.rs
programs/securitize-off-ramp/src/states/liquidity_provider/liquidity_provider_core.rs
programs/securitize-off-ramp/src/states/nav_provider/amm_nav_provider.rs
programs/securitize-off-ramp/src/states/nav_provider/nav_provider_core.rs
programs/securitize-off-ramp/src/states/nav_provider/standard_nav_provider.rs
programs/securitize-off-ramp/src/states/token_type.rs
programs/securitize-off-ramp/src/utils/ed25519.rs
programs/securitize-off-ramp/src/utils/redemption_manager/execute_single_step_redemption.rs
programs/securitize-off-ramp/src/utils/redemption_manager/execute_two_step_redemption.rs
programs/securitize-off-ramp/src/utils/redemption_manager/mod.rs
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

The Securitize Solana On-/Off-Ramp protocol is a regulated real-world asset (RWA) infrastructure system designed to provide compliant entry and exit paths between stablecoin liquidity (USDC) and tokenized assets on Solana. The protocol enables primary subscriptions (issuance), secondary-style swaps, and redemptions for both DS Protocol tokens (regulated RWAs) and frozen-by-default SPL tokens (public equities), serving as the on-chain bridge between fiat-denominated stablecoins and tokenized securities.

The system operates through two core Solana programs built on the Anchor framework:

- **On-ramp:** converts stablecoin deposits into tokenized assets. An investor submits USDC, the program queries an external NAV provider for the current asset rate, calculates the corresponding asset amount after fees, splits the liquidity between a fee collector and a custodian wallet, and supplies asset tokens to the investor via one of three provider mechanisms (allowance-based transfer, direct minting, or SPL Access Control minting).
- **Off-ramp:** performs the inverse. A redeemer submits asset tokens, the program calculates the equivalent liquidity amount at the current NAV rate, handles the asset tokens according to policy (burn or transfer to a designated recipient), and supplies stablecoin liquidity from a delegated allowance provider.

Fee calculation uses ceiling division with a configurable basis-points numerator capped at 50%. Both programs support a two-step transfer mode that routes tokens through an intermediate vault before final delivery.

DS Token flows use the RWA Token suite (asset controller, identity registry, and policy engine).

9 Attestation

Cyfrin attests that it performed a security review of the Solana On/Off Ramp code provided by Securitize.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2026-02-19-cyfrin-securitize-svm-on-off-ramp-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Securitize has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=e6f12f952a166cc49ce135583ebddaa0a0a8845368abb673331b155acfd52403
F=2026-02-19-cyfrin-securitize-svm-on-off-ramp-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.